Optimal Scoring Rules With Random Utility Agents - week 1 progress

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1 Empirical motivation

We assume that the agent does not perfectly maximize expected score but instead optimizes a noisy objective. Under standard theory, any strictly proper scoring rule should elicit identical truthful reports from a risk-neutral expected utility maximizer, so the choice among proper scoring rules would be irrelevant. Experimental evidence contradicts this prediction.

Palfrey and Wang [**?**] elicit probabilistic forecasts from trained observers using three scoring rules: quadratic (proper), logarithmic (proper), and linear (improper). Despite both proper rules being incentive-compatible under the classical model, the two rules produce significantly different forecast distributions. Forecasts under the quadratic rule are substantially more dispersed than those under the logarithmic rule: the average absolute deviation from 50% is 16.4 under the quadratic rule versus 10.2 under the logarithmic rule, and the quadratic distribution stochastically dominates the logarithmic one in extremeness. The authors note that risk aversion under constant relative risk aversion cannot account for this difference, and leave the explanation as an open question.

The random utility model provides a natural explanation. Under the quadratic (Brier) score, $G''(p) = 2$ is constant, while under the logarithmic score, $G''(p) = 1/[p(1-p)]$, which diverges as $p \to 0$ or $p \to 1$. By equation (**??**), the reporting variance at base rate p is approximately $\sigma^2/[G''(p)]^2$. The logarithmic rule therefore imposes a much steeper penalty for deviations near the boundaries of $[0, 1]$, suppressing extreme reports—exactly the pattern observed by Palfrey and Wang.

More broadly, the model belongs to the class of quantal response equilibrium (QRE) models introduced by McKelvey and Palfrey [**?**], in which agents choose actions with probabilities increasing in expected payoff rather than deterministically best-responding. In such models, the shape of the payoff landscape affects the distribution of chosen actions, even when the set of optimal actions is identical across payoff specifications. The extensive experimental support for QRE across strategic settings (see Goeree, Holt, and Palfrey [**?**] for a survey) provides a behavioral foundation for our assumption that the scoring rule's curvature—not merely its properness—determines reporting quality.

2 Setup

Let X be a binary random variable and let $p = \Pr(X = 1)$. The principal's ex ante utility function is given by:

$$u(d, p) = p\, u(d, 1) + (1 - p)\, u(d, 0).$$

However, the principal does not know p. The principal delegates the task of estimating p to an agent. Given the agent's estimate $\hat{p}$, the principal's decision problem is:

$$u(d, \hat{p}) = \hat{p}\, u(d, 1) + (1 - \hat{p})\, u(d, 0).$$

Suppose that there exists a unique optimal decision function $d^*(\hat{p}) \equiv \arg\max_d u(d, \hat{p})$ and that, for all $\hat{p} \in (0, 1)$, $d^*(\hat{p})$ is the unique decision satisfying the first-order condition:

$$\hat{p}\, u'(d^*(\hat{p}), 1) + (1 - \hat{p})\, u'(d^*(\hat{p}), 0) = 0. \tag{1}$$

Hence, the principal's utility when acting optimally is:

$$u(d^*(\hat{p}), p) = p\, u(d^*(\hat{p}), 1) + (1 - p)\, u(d^*(\hat{p}), 0).$$

3 The scoring rule S_u

The principal pays the agent in accordance with a scoring rule S:

$$S(\hat{p}, X) = \begin{cases} S(\hat{p}, 1) & \text{if } X = 1, \\ S(\hat{p}, 0) & \text{if } X = 0. \end{cases}$$

We define the *decision-matched* scoring rule S_u by:

$$S_u(\hat{p}, X) = \begin{cases} u(d^*(\hat{p}), 1) & \text{if } X = 1, \\ u(d^*(\hat{p}), 0) & \text{if } X = 0. \end{cases}$$

Conjecture 1. *The scoring rule S_u is optimal for the principal in the random utility model defined in Section 5, in the sense that it minimizes the expected loss from the agent's reporting error.*

4 Properness of S_u

Proposition 1. *Suppose that $u(d, x)$ is strictly concave in d for each $x \in \{0, 1\}$, and that $\frac{\partial d^*(\hat{p})}{\partial \hat{p}} \neq 0$ for all $\hat{p} \in (0, 1)$. Then S_u is a strictly proper scoring rule.*

Proof. The agent chooses $\hat{p}$ to maximize

$$V_p(\hat{p}) = p \cdot u(d^*(\hat{p}), 1) + (1 - p) \cdot u(d^*(\hat{p}), 0).$$

First-order condition. By the chain rule:

$$V_p'(\hat{p}) = \frac{\partial d^*}{\partial \hat{p}} \Big[p \cdot u'(d^*(\hat{p}), 1) + (1 - p) \cdot u'(d^*(\hat{p}), 0) \Big].$$

Since $\frac{\partial d^*}{\partial \hat{p}} \neq 0$ by assumption, setting $V_p'(\hat{p}) = 0$ gives:

$$p \cdot u'(d^*(\hat{p}), 1) + (1 - p) \cdot u'(d^*(\hat{p}), 0) = 0. \tag{2}$$

Subtracting the principal's first-order condition (1) from (2):

$$(\hat{p} - p) \Big[u'(d^*(\hat{p}), 0) - u'(d^*(\hat{p}), 1) \Big] = 0.$$

Either $\hat{p} = p$, or $u'(d^*(\hat{p}), 1) = u'(d^*(\hat{p}), 0)$. The latter, combined with (1), implies $u'(d^*(\hat{p}), 1) = u'(d^*(\hat{p}), 0) = 0$, meaning d^* maximizes utility in both states simultaneously and the forecast is irrelevant. This is ruled out by the assumption $\frac{\partial d^*}{\partial \hat{p}} \neq 0$. Hence $\hat{p} = p$ is the unique stationary point of V_p.

Second-order condition. Define $B(\hat{p}) = p \cdot u'(d^*(\hat{p}), 1) + (1-p) \cdot u'(d^*(\hat{p}), 0)$, so that $V_p'(\hat{p}) = \frac{\partial d^*}{\partial \hat{p}} \cdot B(\hat{p})$. By the product rule:

$$V_p''(\hat{p}) = \frac{\partial^2 d^*}{\partial \hat{p}^2} \cdot B(\hat{p}) + \frac{\partial d^*}{\partial \hat{p}} \cdot B'(\hat{p}).$$

At $\hat{p} = p$, we have $B(p) = 0$ from (2), so the first term vanishes. Applying the chain rule to B:

$$B'(\hat{p}) = \frac{\partial d^*}{\partial \hat{p}} \Big[p \cdot u''(d^*(\hat{p}), 1) + (1-p) \cdot u''(d^*(\hat{p}), 0) \Big].$$

Therefore:

$$V_p''(p) = \left(\frac{\partial d^*}{\partial \hat{p}} \right)^2 \Big[p \cdot u''(d^*(p), 1) + (1-p) \cdot u''(d^*(p), 0) \Big].$$

The squared derivative is positive. Since $u(d, x)$ is strictly concave in d for each x, we have $u''(d^*(p), 1) < 0$ and $u''(d^*(p), 0) < 0$, so $V_p''(p) < 0$. The unique stationary point $\hat{p} = p$ is a strict maximum, and S_u is strictly proper. □

5 The random utility model

The agent chooses $\hat{p} \in (0, 1)$ to maximize

$$p\, S(\hat{p}, 1) + (1-p)\, S(\hat{p}, 0) + \varepsilon(\hat{p}),$$

where $\varepsilon(\hat{p}) = \theta \hat{p}$ with $\theta \sim N(0, \sigma^2)$. Here θ is a random shock to the agent's evaluation of candidate reports; the linear specification $\varepsilon(\hat{p}) = \theta \hat{p}$ means that higher reports are more (or less) attractive by an amount proportional to θ. (The Gaussian assumption and the linear specification are both stronger than necessary for the main results; see Section 13 for a robustness analysis.)

Let $V_p(\hat{p}) = p \cdot S(\hat{p}, 1) + (1-p) \cdot S(\hat{p}, 0)$ denote the expected score when the true probability is p and the agent reports $\hat{p}$. The agent's first-order condition is:

$$V_p'(\hat{p}) + \theta = 0.$$

A first-order Taylor expansion of V_p' around p gives the approximate solution:

$$\hat{p} \approx p - \frac{\theta}{V_p''(p)}, \tag{3}$$

so that:

$$\mathrm{Var}(\hat{p} - p) \approx \frac{\sigma^2}{[V_p''(p)]^2}. \tag{4}$$

The curvature $V_p''(p)$ is the key object that makes different scoring rules comparable: larger $|V_p''(p)|$ means smaller reporting variance for a given noise level σ^2.

Observations

If $\sigma^2 = 0$, any strictly proper scoring rule yields $\hat{p} = p$. When $\sigma^2 > 0$, different strictly proper scoring rules generally produce different reporting precision because they differ in $V_p''(p)$. Multiplying any scoring rule by a constant $a > 1$ scales $V_p''(p)$ by a, reducing variance by a^2.

The conjecture that S_u is optimal amounts to saying that matching the curvature profile of the scoring rule to the principal's loss function is optimal: S_u concentrates curvature $|V_p''(p)|$ in the regions of p where the principal's decision is most sensitive to the probability estimate.

6 The principal's optimization problem

The analysis so far shows that the curvature of the expected score function controls the agent's reporting precision, and that different strictly proper scoring rules induce different curvature profiles. We now formalize the principal's problem of choosing the optimal scoring rule.

For any scoring rule S, define the *entropy function*

$$G(q) = q\, S(q, 1) + (1 - q)\, S(q, 0), \tag{5}$$

the expected score when the agent's true belief and report both equal q. Note that G differs from V_p: the latter fixes the true probability at p and varies the report, while G evaluates the expected score along the diagonal where report equals truth. One can show using the standard representation of proper scoring rules (see Section 7.1) that $[V_p''(p)]^2 = [G''(p)]^2$ for any strictly proper S, so the variance formula (4) can equivalently be written as

$$\mathrm{Var}(\hat{p} - p) \approx \frac{\sigma^2}{[G''(p)]^2}. \tag{6}$$

6.1 The principal's expected loss

When the agent has true belief p but reports $\hat{p}$, the principal acts with decision $d^*(\hat{p})$ and receives utility $u(d^*(\hat{p}), p)$. The loss relative to knowing p exactly is:

$$L = u(d^*(p), p) - u(d^*(\hat{p}), p).$$

Expanding $u(d^*(\hat{p}), p)$ to second order in $\delta = \hat{p} - p$:

$$\frac{d}{d\hat{p}} u(d^*(\hat{p}), p) = u'(d^*(\hat{p}), p) \cdot (d^*)'(\hat{p}).$$

At $\hat{p} = p$, this vanishes because $u'(d^*(p), p) = 0$ by the principal's first-order condition. The second derivative is:

$$\frac{d^2}{d\hat{p}^2} u(d^*(\hat{p}), p) = u''(d^*(\hat{p}), p) \cdot [(d^*)'(\hat{p})]^2 + u'(d^*(\hat{p}), p) \cdot (d^*)''(\hat{p}).$$

At $\hat{p} = p$, the second term again vanishes, leaving:

$$\left.\frac{d^2}{d\hat{p}^2} u(d^*(\hat{p}), p)\right|_{\hat{p}=p} = u''(d^*(p), p) \cdot [(d^*)'(p)]^2,$$

where $u''(d^*(p), p) = p\, u''(d^*(p), 1) + (1 - p)\, u''(d^*(p), 0) < 0$ by strict concavity. Define the principal's sensitivity:

$$H(p) \equiv u''(d^*(p), p) \cdot [(d^*)'(p)]^2. \tag{7}$$

Note that $H(p) < 0$. The magnitude $|H(p)|$ measures how sensitive the principal's utility is to small errors in the reported probability at base rate p. It is large when the principal's utility is sharply curved at the optimum (large $|u''|$) or when the optimal decision changes rapidly with the probability estimate (large $|(d^*)'|$). The Taylor expansion gives:

$$L \approx -\frac{1}{2} H(p) \cdot \delta^2,$$

which is positive since $H(p) < 0$.

6.2 Expected loss under the random utility model

From (6), the expected squared error for scoring rule S is $\mathbb{E}[\delta^2] \approx \sigma^2/[G''(p)]^2$. Therefore the expected loss at a given p is:

$$\mathbb{E}[L \mid p] \approx -\frac{\sigma^2 \, H(p)}{2[G''(p)]^2}. \tag{8}$$

6.3 The scaling problem

Replacing S with aS for $a > 1$ scales $G''(p)$ by a, reducing the expected loss by a factor of a^2. As $a \to \infty$, the expected loss vanishes for any strictly proper scoring rule. Therefore, comparing scoring rules without constraining the scale of payments is meaningless: the principal can always improve by simply paying the agent more.

6.4 Net utility formulation

To obtain a well-posed optimization problem, we introduce a cost $\alpha > 0$ per unit of expected payment to the agent. When the agent reports approximately truthfully, the expected payment at belief p is $G(p)$. The principal's net expected utility at a given p is the negative of the expected loss minus the cost of payment:

$$V(p) = -\mathbb{E}[L \mid p] - \alpha \, G(p) \approx \frac{\sigma^2 \, H(p)}{2[G''(p)]^2} - \alpha \, G(p).$$

Note that $H(p) < 0$, so the first term is negative (a loss), and the second term is also negative (a cost). Integrating over the principal's prior $f(p)$ on the true probability, the principal's problem is:

$$\max_{S \in \mathcal{S}} \int_0^1 \left[\frac{\sigma^2 \, H(p)}{2[G''(p)]^2} - \alpha \, G(p) \right] f(p) \, dp, \tag{9}$$

where $\mathcal{S}$ denotes the set of strictly proper scoring rules. The tradeoff is clear: increasing the curvature $|G''(p)|$ reduces the loss from reporting error, but requires higher expected payments $G(p)$.

7 Optimization under limited liability

The formulation (9) is not well-posed without additional constraints: adding a function of x alone to S preserves properness and G'' but shifts expected payments arbitrarily. We impose limited liability: $S(\hat{p}, x) \geq 0$ for all $\hat{p} \in (0, 1)$ and $x \in \{0, 1\}$.

7.1 Reducing the constraint

The standard representation of proper scoring rules expresses S in terms of the entropy function G:

$$S(\hat{p},1) = G(\hat{p}) + (1-\hat{p})\,G'(\hat{p}), \qquad S(\hat{p},0) = G(\hat{p}) - \hat{p}\,G'(\hat{p}). \tag{10}$$

To derive this, recall that $G(q) = q\,S(q,1) + (1-q)\,S(q,0)$. Differentiating with respect to q:

$$G'(q) = S(q,1) - S(q,0) + q\,S_1(q,1) + (1-q)\,S_1(q,0),$$

where S_1 denotes the partial derivative of S with respect to its first argument. By properness, $V_p(\hat{p}) = p\,S(\hat{p},1) + (1-p)\,S(\hat{p},0)$ is maximized at $\hat{p} = p$ for all p, which requires $p\,S_1(p,1) + (1-p)\,S_1(p,0) = 0$ for all p. Evaluating at $p = q$, the last two terms vanish and we obtain $G'(q) = S(q,1) - S(q,0)$. Combining with $G(q) = q\,S(q,1) + (1-q)\,S(q,0)$ and solving the resulting 2×2 system yields (10).

Differentiating the standard representation with respect to $\hat{p}$:

$$\frac{\partial}{\partial\hat{p}} S(\hat{p},1) = (1-\hat{p})\,G''(\hat{p}), \qquad \frac{\partial}{\partial\hat{p}} S(\hat{p},0) = -\hat{p}\,G''(\hat{p}).$$

For a strictly proper scoring rule, G is strictly convex, so $G''(\hat{p}) > 0$. Therefore $S(\hat{p},1)$ is increasing in $\hat{p}$ and $S(\hat{p},0)$ is decreasing in $\hat{p}$. Their respective minima over $\hat{p} \in [0,1]$ are:

$$\min_{\hat{p}} S(\hat{p},1) = S(0,1) = G(0) + G'(0), \qquad \min_{\hat{p}} S(\hat{p},0) = S(1,0) = G(1) - G'(1).$$

The infinite-dimensional limited liability constraint therefore reduces to:

$$G(0) + G'(0) \geq 0, \qquad G(1) - G'(1) \geq 0. \tag{11}$$

Remark 1 (Connection between V_p'' and G'')**.** *From the standard representation, one can verify that $V_p(\hat{p}) = G(\hat{p}) + (p-\hat{p})\,G'(\hat{p})$. Differentiating twice with respect to $\hat{p}$ and evaluating at $\hat{p} = p$ gives $V_p''(p) = -G''(p)$. Since all formulas in the principal's problem* (9) *involve $[G''(p)]^2$, this sign difference does not affect the optimization.*

7.2 Both constraints bind

Adding an affine function $a + bp$ to G does not change G'', so the loss term in (9) is unaffected. The expected payment $G(p)$ shifts by $a + bp$, which is a positive cost for the principal. Since a and b provide two degrees of freedom and there are two inequality constraints, the principal minimizes payments by choosing a and b so that both constraints bind:

$$G(0) + G'(0) = 0, \qquad G(1) - G'(1) = 0. \tag{12}$$

7.3 Green's function representation

The loss term in (9) depends directly on $g = G'' > 0$, but the payment term depends on G. To optimize over g, we express G as a linear functional of g using the boundary conditions (12).

Starting from $G''(p) = g(p)$, integrate once from 0 to p:

$$G'(p) = G'(0) + \int_0^p g(t)\,dt.$$

Integrate again from 0 to p. The left side gives $G(p) - G(0)$. On the right, the double integral $\int_0^p \int_0^s g(t)\,dt\,ds$ can be evaluated by swapping the order of integration: writing $t \le s \le p$ equivalently as $t \le p$ and $t \le s \le p$, we obtain $\int_0^p (p-t)\,g(t)\,dt$. Therefore:

$$G(p) = G(0) + G'(0)\,p + \int_0^p (p-t)\,g(t)\,dt. \tag{13}$$

We now determine $G(0)$ and $G'(0)$ from the boundary conditions (12).

First boundary condition. $G(0) + G'(0) = 0$ gives:

$$G(0) = -G'(0). \tag{14}$$

Second boundary condition. Evaluating (13) at $p = 1$:

$$G(1) = G(0) + G'(0) + \int_0^1 (1-t)\,g(t)\,dt = \int_0^1 (1-t)\,g(t)\,dt,$$

where the first two terms cancel by (14). From the first integral, $G'(1) = G'(0) + \int_0^1 g(t)\,dt$. The condition $G(1) = G'(1)$ becomes:

$$\int_0^1 (1-t)\,g(t)\,dt = G'(0) + \int_0^1 g(t)\,dt.$$

Solving for $G'(0)$:

$$G'(0) = \int_0^1 (1-t)\,g(t)\,dt - \int_0^1 g(t)\,dt = -\int_0^1 t\,g(t)\,dt.$$

Combined with (14): $G(0) = \int_0^1 t\,g(t)\,dt$.

Assembling G. Substituting into (13):

$$\begin{aligned} G(p) &= \int_0^1 t\,g(t)\,dt - p\int_0^1 t\,g(t)\,dt + \int_0^p (p-t)\,g(t)\,dt \\ &= (1-p)\int_0^1 t\,g(t)\,dt + \int_0^p (p-t)\,g(t)\,dt. \end{aligned} \tag{15}$$

To combine into a single integral, split the first integral at $t = p$:

$$(1-p)\int_0^1 t\,g(t)\,dt = (1-p)\int_0^p t\,g(t)\,dt + (1-p)\int_p^1 t\,g(t)\,dt.$$

For $t \le p$, collecting terms from both integrals in (15):

$$(1-p)\,t + (p-t) = t - pt + p - t = p - pt = p(1-t).$$

For $t > p$, only the first integral contributes: the coefficient is $(1-p)\,t = t(1-p)$. Therefore:

$$G(p) = \int_0^1 K(p,t)\,g(t)\,dt, \tag{16}$$

where:

$$K(p,t) = \begin{cases} p(1-t), & t \le p, \\ t(1-p), & t > p. \end{cases} \tag{17}$$

Note that $K(p,t) = K(t,p)$, i.e. the kernel is symmetric. Since $g > 0$ and $K \ge 0$ on $(0,1)^2$, we have $G(p) \ge 0$ as required.

7.4 Reformulation in terms of g

We now rewrite the principal's objective (9) entirely in terms of $g(p) = G''(p) > 0$. The loss term already depends only on $g(p)$ at each p:

$$\text{Loss term} = \int_0^1 \frac{\sigma^2 H(p) f(p)}{2\, g(p)^2}\, dp.$$

The payment term is more involved. Substituting $G(p) = \int_0^1 K(p,t)\, g(t)\, dt$:

$$\text{Payment term} = \int_0^1 \alpha\, f(p) \left[\int_0^1 K(p,t)\, g(t)\, dt\right] dp.$$

This is a double integral where g appears inside the inner integral evaluated at t, not at p: the choice of g at one point affects the payment cost at all other points. To decouple this, we swap the order of integration (justified since all integrands are continuous on $[0,1]^2$):

$$\int_0^1 \alpha\, f(p) \left[\int_0^1 K(p,t)\, g(t)\, dt\right] dp = \int_0^1 \alpha\, g(t) \left[\int_0^1 K(p,t)\, f(p)\, dp\right] dt.$$

The inner integral on the right is a fixed function of t. We define:

$$\phi(p) \equiv \int_0^1 K(t,p)\, f(t)\, dt. \tag{18}$$

Splitting the integral at $p = t$ and using (17) (noting that in the first integral $p \le t$, so $K(p,t) = t(1-p)$; in the second $p > t$, so $K(p,t) = p(1-t)$):

$$\phi(t) = t \int_0^t (1-p)\, f(p)\, dp + (1-t) \int_t^1 p\, f(p)\, dp. \tag{19}$$

Since $K \ge 0$ and $f \ge 0$, we have $\phi(t) > 0$ for $t \in (0,1)$.

The magnitude of $\phi(t)$ measures the cost of curvature at base rate t: increasing $g(t)$ raises the expected payment $G(p) = \int K(p,t)\, g(t)\, dt$ across all base rates p, and $\phi(t)$ is the prior-weighted total of that increase.

Renaming the dummy variable from t back to p, the full objective becomes:

$$\int_0^1 \left[\frac{\sigma^2 H(p) f(p)}{2\, g(p)^2} - \alpha\, \phi(p)\, g(p)\right] dp. \tag{20}$$

This is *separable* in g: the integrand at each p depends only on $g(p)$. Therefore, maximizing the integral reduces to maximizing the integrand independently at each p.

7.5 Pointwise optimization

At each p, we maximize the integrand as a function of $g > 0$:

$$h(g) = \frac{\sigma^2 H(p) f(p)}{2\, g^2} - \alpha\, \phi(p)\, g.$$

Recall that $H(p) < 0$, $f(p) > 0$, and $\phi(p) > 0$.

First derivative. Using $\frac{d}{dg} g^{-2} = -2g^{-3}$:

$$h'(g) = \frac{-\sigma^2 H(p) f(p)}{g^3} - \alpha \phi(p).$$

The first term is positive (since $H(p) < 0$ and $g > 0$), and the second is negative. Setting $h'(g) = 0$ and solving:

$$g^3 = \frac{-\sigma^2 H(p) f(p)}{\alpha \phi(p)} = \frac{\sigma^2 |H(p)| f(p)}{\alpha \phi(p)}. \tag{21}$$

The right-hand side is positive, so $g^* > 0$ as required.

Second derivative. Using $\frac{d}{dg} g^{-3} = -3g^{-4}$:

$$h''(g) = \frac{3\sigma^2 H(p) f(p)}{g^4}.$$

Since $H(p) < 0$, we have $h''(g) < 0$ for all $g > 0$: h is globally concave, and g^* is the unique maximum.

The optimal curvature is therefore:

$$\boxed{G''(p)^* = \left(\frac{\sigma^2 |H(p)| f(p)}{\alpha \phi(p)} \right)^{1/3},} \tag{22}$$

where $H(p)$ is the principal's sensitivity (7) and $\phi(p)$ is the cost of curvature (18).

The optimal scoring rule concentrates curvature where errors are costly ($|H(p)|$ large), where the base rate is likely ($f(p)$ large), and where curvature is cheap ($\phi(p)$ small).

7.6 Existence of the optimal scoring rule

The pointwise optimization yields an optimal curvature function $g^*(p)$, but we must verify that a strictly proper scoring rule with this curvature actually exists. Specifically, we need to show that integrating g^* twice and imposing the boundary conditions (12) produces a valid entropy function G^*.

By the Green's function representation (16), define

$$G^*(p) = \int_0^1 K(p,t)\, g^*(t)\, dt,$$

where K is the kernel (17). This integral is well-defined and finite for all $p \in [0,1]$ provided g^* is integrable on $(0,1)$, which holds whenever $|H(p)| f(p)/\phi(p)$ is bounded (a condition guaranteed by the assumptions in Section 7: $\phi(p) > 0$ on $(0,1)$ and $|H(p)| f(p)$ bounded).

We verify that G^* satisfies all requirements for a valid entropy function of a strictly proper scoring rule with limited liability.

1. **Strict convexity.** By construction, $(G^*)''(p) = g^*(p) > 0$ for all $p \in (0,1)$, so G^* is strictly convex.

2. **Boundary conditions.** As shown in Section 7.3, any G constructed via (16) automatically satisfies $G(0) + G'(0) = 0$ and $G(1) - G'(1) = 0$. Indeed, from the derivation: $G^*(0) = \int_0^1 K(0,t)\, g^*(t)\, dt = 0$ since $K(0,t) = 0$ for all t. From (13), $(G^*)'(0) = -\int_0^1 t\, g^*(t)\, dt$ and $G^*(0) = \int_0^1 t\, g^*(t)\, dt$, giving $G^*(0) + (G^*)'(0) = 0$. The second boundary condition follows analogously.

3. **Limited liability.** Since the boundary conditions bind with equality, the reduced constraints (11) hold: $\min_{\hat{p}} S(\hat{p}, 1) = G^*(0) + (G^*)'(0) = 0 \geq 0$ and $\min_{\hat{p}} S(\hat{p}, 0) = G^*(1) - (G^*)'(1) = 0 \geq 0$. Therefore $S(\hat{p}, x) \geq 0$ for all $\hat{p} \in (0, 1)$ and $x \in \{0, 1\}$.

4. **Non-negative expected payments.** Since $K(p, t) \geq 0$ and $g^*(t) > 0$, we have $G^*(p) \geq 0$ for all $p \in [0, 1]$, with $G^*(p) > 0$ for $p \in (0, 1)$.

The scoring rule corresponding to G^* is then given by the standard representation (10):

$$S^*(\hat{p}, 1) = G^*(\hat{p}) + (1 - \hat{p})\,(G^*)'(\hat{p}), \qquad S^*(\hat{p}, 0) = G^*(\hat{p}) - \hat{p}\,(G^*)'(\hat{p}).$$

This is a strictly proper scoring rule satisfying limited liability, confirming that the optimal curvature (22) is attained by an admissible scoring rule.

7.7 When is S_u optimal?

For the decision-matched scoring rule S_u, the entropy function is

$$G(q) = q \cdot u(d^*(q), 1) + (1 - q) \cdot u(d^*(q), 0) = u(d^*(q), q).$$

By a direct computation analogous to Section 6.1, $G''(p) = -H(p) > 0$ (see Remark 1). Substituting $G''(p) = |H(p)|$ into (22):

$$|H(p)|^3 = \frac{\sigma^2 |H(p)| f(p)}{\alpha\,\phi(p)} \implies |H(p)|^2 = \frac{\sigma^2 f(p)}{\alpha\,\phi(p)}.$$

This must hold for all p simultaneously with fixed σ^2 and α. Equivalently, S_u is optimal if and only if $|H(p)|^2\,\phi(p)/f(p)$ is constant in p. For a generic choice of u and f, this will not hold, so S_u is not generically optimal.

Counterexample to Conjecture 1. Take $u(d, x) = -(d - x)^2$ and $f(p) = 1$ (uniform prior). Then $d^*(p) = p$, $H(p) = -2$, and S_u coincides with the Brier score (up to an additive shift to satisfy limited liability), which has constant curvature $G''(p) = 2$.

We compute $\phi(t)$ from (19) with $f \equiv 1$:

$$\begin{aligned}
\phi(t) &= t \int_0^t (1 - p)\,dp + (1 - t) \int_t^1 p\,dp \\
&= t \left[p - \frac{p^2}{2} \right]_0^t + (1 - t) \left[\frac{p^2}{2} \right]_t^1 \\
&= t \left(t - \frac{t^2}{2} \right) + (1 - t) \left(\frac{1 - t^2}{2} \right).
\end{aligned}$$

Expanding:

$$\begin{aligned}
\phi(t) &= t^2 - \frac{t^3}{2} + \frac{1 - t^2}{2} - \frac{t - t^3}{2} \\
&= t^2 - \frac{t^3}{2} + \frac{1}{2} - \frac{t^2}{2} - \frac{t}{2} + \frac{t^3}{2} \\
&= \frac{t^2 - t + 1}{2}.
\end{aligned}$$

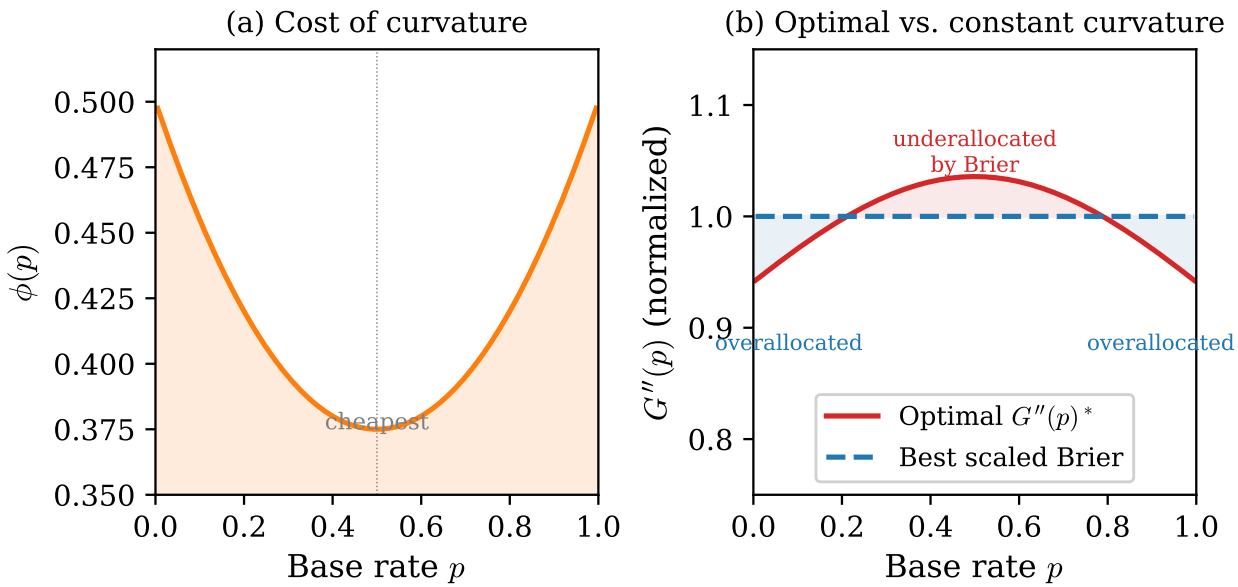

Figure 1: Counterexample with $u(d, x) = -(d - x)^2$ and uniform prior. (a) The cost of curvature $\phi(p)$ is U-shaped: adding curvature near $p = 1/2$ is cheapest under limited liability. (b) Since the principal's sensitivity $|H(p)|$ is constant, the optimal rule concentrates curvature where it is cheap. The Brier score's constant curvature overallocates near the boundaries and underallocates near $p = 1/2$.

The optimality condition requires $|H(p)|^2 \, \phi(p)/f(p) = 4 \cdot (p^2 - p + 1)/2 = 2(p^2 - p + 1)$ to be constant. Since $p^2 - p + 1$ has a minimum of $3/4$ at $p = 1/2$ and equals 1 at the boundaries, this fails.

The optimal curvature is:

$$G''(p)^* = \left(\frac{2\sigma^2}{\alpha \cdot (p^2 - p + 1)/2} \right)^{1/3} = \left(\frac{4\sigma^2}{\alpha \, (p^2 - p + 1)} \right)^{1/3}.$$

This is maximized at $p = 1/2$ (where ϕ is smallest, so curvature is cheapest) and minimized at $p \in \{0, 1\}$ (where ϕ is largest). The Brier score spreads curvature uniformly, under-allocating it near $p = 1/2$ and over-allocating it near the boundaries relative to the optimum.

7.8 Boundary behavior

Under the limited liability constraints (12), the cost of curvature $\phi(p)$ remains bounded away from zero at the boundaries: $\phi(0) = \int_0^1 p \, f(p) \, dp = \mathbb{E}[p]$ and $\phi(1) = \int_0^1 (1 - p) \, f(p) \, dp = \mathbb{E}[1 - p]$, both of which are strictly positive whenever the prior f does not concentrate all mass at a single point. Therefore, provided $|H(p)| \, f(p)$ remains bounded as $p \to 0$ or $p \to 1$, the optimal curvature (22) stays finite at the boundaries, and the scoring rule has bounded payments.

8 The Boundary problem

Under the random utility model, the agent selects a report $\hat{p}$ to maximize their expected score perturbed by linear noise:

$$\max_{\hat{p}} V_p(\hat{p}) + \theta\hat{p}, \quad \theta \sim N(0, \sigma^2)$$

Let $\hat{p}_{exact}$ denote the exact unconstrained maximizer, i.e. the solution to $V_p'(\hat{p}) + \theta = 0$. A first-order Taylor expansion around p gives the approximation $\hat{p}_{exact} \approx p + \theta/G''(p)$.

Because the Gaussian noise θ has infinite support, there is a strictly positive probability that $\hat{p}_{exact} \notin [0, 1]$ for any base rate $p \in (0, 1)$ and noise level $\sigma > 0$. This violates the fundamental requirement that the agent's report must be a valid probability.

8.1 Solution: Clipping

To resolve this, we explicitly constrain the agent's action space to the valid probability interval $[0, 1]$. Because S is a strictly proper scoring rule, the expected score $V_p(\hat{p})$ is strictly concave in $\hat{p}$. The addition of the linear noise term $\theta\hat{p}$ preserves this strict concavity.

For a strictly concave objective function maximized over a closed interval, if the exact unconstrained maximum $\hat{p}_{exact}$ falls outside the interval, the constrained maximum is the nearest boundary point. The agent's actual report is therefore the clipped value:

$$\hat{p}_c = \max(0, \min(1, \hat{p}_{exact})) \tag{23}$$

We show that among all valid reports $\hat{p} \in [0, 1]$, the clipped value $\hat{p}_c$ is optimal for the agent, and that it yields a lower loss for the principal than any report further from the truth.

The principal's utility when acting on a report $\hat{p} \in [0, 1]$ is $W(\hat{p}) \equiv u(d^*(\hat{p}), p)$. By definition of the decision-matched scoring rule, this is exactly the expected score function $V_p(\hat{p})$ analyzed in Proposition 1. As shown in the proof of Proposition 1, $W(\hat{p})$ has a unique stationary point at $\hat{p} = p$, which is a strict local maximum. Since W is continuously differentiable on $(0, 1)$ and W' has no other zeros, the intermediate value theorem implies that W' cannot change sign on $(0, p)$ or on $(p, 1)$. Combined with $W'(p^-) > 0$ and $W'(p^+) < 0$ (from the local maximum), it follows that $W(\hat{p})$ is strictly increasing on $[0, p]$ and strictly decreasing on $[p, 1]$.

When $\hat{p}_{exact} \notin [0, 1]$, the clipped report $\hat{p}_c$ is the nearest boundary point (0 or 1). Since $p \in (0, 1)$, the boundary point lies strictly between p and $\hat{p}_{exact}$, so $|\hat{p}_c - p| < |\hat{p}_{exact} - p|$. By the monotonicity of W, the boundary report yields strictly higher principal utility than any valid report further from p. In particular, $\hat{p}_c$ minimizes the principal's loss $L(\hat{p}) = u(d^*(p), p) - u(d^*(\hat{p}), p)$ among all reports on the same side of p as $\hat{p}_{exact}$.

We now show that clipping does not affect the optimal curvature derived in the unconstrained analysis. The principal's loss under the quadratic Taylor approximation is $L(\hat{p}) \approx \frac{1}{2}|H(p)|(\hat{p} - p)^2$, which is strictly increasing in $|\hat{p} - p|$. Since clipping satisfies $|\hat{p}_c - p| \leq |\hat{p}_{exact} - p|$ for every realization of θ, with strict inequality whenever $\hat{p}_{exact} \notin [0, 1]$, the clipped loss is bounded above by the unconstrained loss:

$$L(\hat{p}_c) \leq L(\hat{p}_{exact}) \approx \frac{1}{2}|H(p)| \cdot \frac{\theta^2}{[G''(p)]^2}. \tag{24}$$

Taking expectations, the unconstrained expected loss $\frac{\sigma^2|H(p)|}{2[G''(p)]^2}$ is an upper bound on the true expected loss under clipping. Since this bound has the same functional dependence on $G''(p)$ as the exact clipped loss—both are decreasing in $G''(p)$ for fixed p—the curvature $G''(p)^*$ that minimizes the unconstrained objective also minimizes the upper bound on the clipped objective. The optimal curvature (22) therefore remains valid in the clipped model.

Note that while the unconstrained objective overestimates the level of expected loss, the overestimation is larger for lower-curvature rules (which produce more clipping). This means the true gain from the optimal rule over a suboptimal alternative is, if anything, understated by the unconstrained analysis.

9 The subcontracting problem

The optimal curvature formula (22) applies to any utility function $u(d, x)$: one computes $H(p)$ and plugs in. We now consider a natural special case. Suppose the principal is herself facing a forecasting problem: she is scored according to a strictly proper scoring rule S_0 and delegates the estimation of p to a sub-agent. This setting lets us analyze the expected gain for the principal from choosing particular scoring rules.

9.1 Setup

The principal faces a strictly proper scoring rule S_0 with entropy function G_0. Her gross payoff from reporting d to the meta-principal when the outcome is x is $S_0(d, x)$. The cost of paying the sub-agent is a separate expenditure, already captured by the $\alpha\, G(p)$ term in the principal's objective (9). The principal's utility function in the sense of Section 6 is therefore $u(d, x) = S_0(d, x)$. Since S_0 is strictly proper, the expected payoff $p\, S_0(d, 1) + (1 - p)\, S_0(d, 0)$ is uniquely maximized at $d = p$, so $d^*(\hat{p}) = \hat{p}$ and $(d^*)'(\hat{p}) = 1$.

9.2 Principal's sensitivity

Since $(d^*)'(p) = 1$, the principal's sensitivity (7) simplifies to:

$$H(p) = u''(d^*(p), p) \cdot 1^2 = \frac{\partial^2}{\partial d^2}\Big[p\, S_0(d, 1) + (1 - p)\, S_0(d, 0)\Big]\Big|_{d=p}.$$

The expression in brackets is the principal's expected payoff from reporting d when the true probability is p. This has the same form as the expected score function analyzed in Remark 1, with S_0 playing the role of S and G_0 playing the role of G. Applying that remark:

$$H(p) = -G_0''(p), \qquad |H(p)| = G_0''(p). \tag{25}$$

The principal's sensitivity to reporting errors at base rate p is determined by the curvature of the scoring rule she faces.

9.3 Optimal curvature for the sub-agent

Substituting $|H(p)| = G_0''(p)$ into (22):

$$G''(p)^* = \left(\frac{\sigma^2\, G_0''(p)\, f(p)}{\alpha\, \phi(p)}\right)^{1/3}. \tag{26}$$

The optimal curvature for the sub-agent scales as $[G_0''(p)]^{1/3}$, not $G_0''(p)$. This means the optimal rule *compresses* the curvature profile of S_0: it retains the qualitative shape (more curvature where G_0'' is large) but flattens it toward uniformity. Where S_0 has high curvature, the optimal rule has proportionally less than S_0; where S_0 has low curvature, the optimal rule has proportionally more.

In particular, the optimal rule for the sub-agent generically differs from S_0 itself. (Note that S_0 coincides with the decision-matched rule S_u in this setting, since $S_u(\hat{p}, x) = u(d^*(\hat{p}), x) = S_0(\hat{p}, x)$. The generic suboptimality of S_0 is therefore a special case of the result in Section 7 that S_u is not generically optimal.)

9.4 Examples

Brier score. If the principal faces the Brier score, $G_0''(p) = 2$ (constant), and $|H(p)| = 2$. The optimal curvature for the sub-agent is $G''(p)^* = (2\sigma^2 f(p)/[\alpha\,\phi(p)])^{1/3}$. With uniform prior, this reduces to $(4\sigma^2/[\alpha(p^2 - p + 1)])^{1/3}$, which is the counterexample from Section 7. The Brier score's constant curvature is suboptimal because $\phi(p)$ is not constant: curvature is cheaper near $p = 1/2$ than at the boundaries.

Log score. If the principal faces the log score, $G_0(p) = p \ln p + (1-p)\ln(1-p)$ and $G_0''(p) = 1/[p(1-p)]$. The sensitivity $|H(p)| = 1/[p(1-p)]$ is large near the boundaries, reflecting the log score's sharp penalties for confident wrong forecasts. The optimal sub-agent curvature is:

$$G''(p)^* = \left(\frac{\sigma^2\, f(p)}{\alpha\,\phi(p)\, p(1-p)}\right)^{1/3}.$$

With uniform prior, this diverges as $p \to 0$ or $p \to 1$, but more slowly than $G_0''(p) = 1/[p(1-p)]$: the optimal curvature scales as $[p(1-p)]^{-1/3}$ versus $[p(1-p)]^{-1}$. The principal who faces the log score should give the sub-agent a rule with less extreme curvature near the boundaries.

9.5 Efficiency of the pass-through rule

A natural benchmark is the *pass-through rule*: the principal assigns the sub-agent a scaled version of her own scoring rule S_0. Since scaling S_0 by $a > 0$ scales G'' by a, the pass-through rule has curvature $g(p) = a \cdot G_0''(p)$ for some $a > 0$ chosen optimally. We compare the principal's net utility under the pass-through to that under the pointwise-optimal curvature (26).

Write $B(p) = G_0''(p) = |H(p)|$ for the curvature of S_0. The principal's net utility under curvature g is (from (20)):

$$V(g) = \int_0^1 \left[-\frac{\sigma^2 B(p)\, f(p)}{2\, g(p)^2} - \alpha\,\phi(p)\, g(p)\right] dp.$$

Optimal rule. Substituting the pointwise-optimal $g^*(p) = (\sigma^2 B(p) f(p)/[\alpha\,\phi(p)])^{1/3}$ into the integrand and simplifying:

$$V^* = -\tfrac{3}{2}\,(\sigma^2\alpha^2)^{1/3} \int_0^1 (B(p)\, f(p))^{1/3}\, \phi(p)^{2/3}\, dp. \tag{27}$$

Pass-through rule. With $g(p) = a\, B(p)$, the objective separates into two terms that depend on a as $1/a^2$ and a respectively:

$$V(a) = -\frac{\sigma^2}{2a^2} \underbrace{\int_0^1 \frac{f(p)}{B(p)}\, dp}_{I_1} - \alpha\, a \underbrace{\int_0^1 \phi(p)\, B(p)\, dp}_{I_2}.$$

Optimizing over $a > 0$ yields $a^* = (\sigma^2 I_1/(\alpha\, I_2))^{1/3}$ and

$$V_{\text{pass}} = -\tfrac{3}{2}\,(\sigma^2\alpha^2)^{1/3}\, I_1^{1/3}\, I_2^{2/3}. \tag{28}$$

The efficiency ratio. Since both V^* and V_{pass} are negative, and V^* is closer to zero, the ratio

$$\eta \equiv \frac{V^*}{V_{\text{pass}}} = \frac{\displaystyle\int_0^1 (B(p)\, f(p))^{1/3}\, \phi(p)^{2/3}\, dp}{\left(\displaystyle\int_0^1 \frac{f(p)}{B(p)}\, dp\right)^{1/3} \left(\displaystyle\int_0^1 \phi(p)\, B(p)\, dp\right)^{2/3}} \tag{29}$$

lies in $(0, 1]$, and the fractional improvement from using the optimal rule is $1 - \eta$. Three properties are immediate.

Parameter-free. The ratio η depends only on the shape functions B, f, and ϕ — not on σ^2 or α. The percentage gain from optimization is the same regardless of the noise level or the cost of payments.

Hölder bound. Apply Hölder's inequality with exponents $(3, \frac{3}{2})$ to $u = f/B$ and $v = \phi\, B$:

$$\int_0^1 (f/B)^{1/3}\, (\phi\, B)^{2/3}\, dp \;\le\; \left(\int_0^1 \frac{f}{B}\, dp\right)^{1/3} \left(\int_0^1 \phi\, B\, dp\right)^{2/3} = I_1^{1/3}\, I_2^{2/3}.$$

The left-hand side equals $\int (Bf)^{1/3}\, \phi^{2/3}\, dp$ (the numerator of η), confirming $\eta \le 1$.

Equality condition. Hölder's inequality is tight if and only if $f/B \propto \phi\, B$, i.e.,

$$\frac{B(p)^2\, \phi(p)}{f(p)} = \frac{|H(p)|^2\, \phi(p)}{f(p)} = \text{const}.$$

This is precisely the condition for $S_u = S_0$ to be optimal (Section 7).

Numerical examples. Table 1 evaluates η for different combinations of S_0 and the prior f. For the Brier score, $B(p) = 2$; for the spherical score, $B(p) = [p^2 + (1-p)^2]^{-3/2}$, which peaks at $p = 1/2$ and equals 1 at the boundaries.

Table 1: Efficiency η of the pass-through rule for different scoring rules and priors.

S_0	Prior f	η	Gain $1 - \eta$
Brier	Uniform	0.999	0.1%
Brier	Beta$(2, 2)$	0.946	5.4%
Brier	Beta$(5, 5)$	0.780	22.0%
Brier	Beta$(0.5, 0.5)$	0.970	3.0%
Spherical	Uniform	0.966	3.4%
Spherical	Beta$(2, 2)$	0.992	0.8%
Spherical	Beta$(5, 5)$	0.896	10.4%
Spherical	Beta$(0.5, 0.5)$	0.868	13.2%

Two patterns stand out. First, the Brier score with a uniform prior — the most commonly studied case — is nearly optimal (0.1% gain), but this is the exception. With a Beta$(5, 5)$ prior, the gain rises to 22%: the Brier score wastes curvature at the boundaries where the prior places negligible mass, while the optimal rule concentrates curvature near $p = 1/2$ where both $f(p)$ and $1/\phi(p)$ are large.

Second, the spherical score is near-optimal for Beta$(2, 2)$ (0.8% gain) but not for Beta$(0.5, 0.5)$ (13.2% gain). Its curvature peaks at $p = 1/2$ and is lowest at the boundaries, which matches a prior concentrated in the middle but conflicts with one concentrated at the extremes. This illustrates the general principle: the pass-through rule is efficient when its curvature profile $B(p)$ happens to satisfy $B(p)^2\, \phi(p)/f(p) \approx$ const, which depends on the prior.

10 Log-odds reparametrization

A limitation of the probability-space model is that the optimal report $\hat{p} = p - \theta/V_p''(p)$ can fall outside $(0,1)$ for any $\sigma^2 > 0$, since θ has unbounded support. A natural fix is to reparametrize the agent's choice in log-odds: let $\hat{\ell} = \text{logit}(\hat{p}) \in \mathbb{R}$, and suppose the agent maximizes

$$\tilde{V}_p(\hat{\ell}) + \theta\hat{\ell}, \qquad \hat{\ell} \in \mathbb{R}, \quad \theta \sim N(0, \sigma^2),$$

where $\tilde{V}_p(\hat{\ell}) = V_p(\sigma(\hat{\ell}))$ and σ is the logistic function. Since $\hat{\ell}$ ranges over $\mathbb{R}$, the reported probability $\hat{p} = \sigma(\hat{\ell}^*)$ is automatically in $(0,1)$ for every noise realization.

This is not merely a change of variables but a different model: the agent now faces additive noise in log-odds rather than in probability space. The first-order condition $\tilde{V}_p'(\hat{\ell}) + \theta = 0$ and a Taylor expansion around the truthful log-odds $\ell = \text{logit}(p)$ yield the reporting variance

$$\text{Var}(\hat{p} - p) \approx \frac{\sigma^2}{G''(p)^2 \cdot p^2(1-p)^2}, \tag{30}$$

which differs from the probability-space formula (6) by a factor of $1/[p^2(1-p)^2]$. The resulting optimal curvature is $G''(p)^*_{\text{lo}} = G''(p)^*_{\text{prob}}/[p^2(1-p)^2]^{1/3}$, which diverges at the boundaries and demands more curvature everywhere than the probability-space solution.

10.1 Failure of the log-odds model for bounded scoring rules

However, the log-odds model has a fundamental defect for bounded scoring rules that does not arise in the probability-space model. The issue is that the ratio of noise to incentive gradient is unbounded near the boundaries for any $\sigma > 0$.

In both models, the agent's first-order condition equates the incentive gradient to the noise: the scoring rule provides a "restoring force" that pulls the agent toward the truthful report, while the noise pushes the agent away. The models differ in how strong this restoring force can be.

Probability-space model. The agent maximizes $V_p(\hat{p}) + \theta\hat{p}$, giving FOC $V_p'(\hat{p}) + \theta = 0$. The incentive gradient $V_p'(\hat{p})$ is defined on all of $(0,1)$ and, for a proper scoring rule, grows in magnitude as $\hat{p}$ moves away from p: truthful reporting is a global maximum of V_p, so the gradient becomes increasingly negative (positive) as $\hat{p}$ overshoots (undershoots) p. For the Brier score, $V_p'(\hat{p}) = 2(p - \hat{p})$, which is linear and unbounded on $\mathbb{R}$. Every noise realization θ has a unique matching report $\hat{p} = p + \theta/2$. The "support problem" — $\hat{p}$ falling outside $(0,1)$ — occurs only when $|\theta| > 2\min(p, 1-p)$, i.e., when the noise exceeds the distance from p to the nearest boundary scaled by $|V_p''(p)| = 2$. At $p = 0.05$ with $\sigma = 0.01$, this requires $|\theta| > 0.1 = 10\sigma$, which has probability $2\Phi(-10) < 10^{-23}$.

Log-odds model. The agent maximizes $\tilde{V}_p(\hat{\ell}) + \theta\hat{\ell}$, giving FOC $\tilde{V}_p'(\hat{\ell}) + \theta = 0$. By the chain rule, $\tilde{V}_p'(\hat{\ell}) = V_p'(\sigma(\hat{\ell})) \cdot \sigma'(\hat{\ell})$. The logistic derivative $\sigma'(\hat{\ell}) = \sigma(\hat{\ell})(1 - \sigma(\hat{\ell})) \to 0$ as $\hat{\ell} \to \pm\infty$, so for any bounded scoring rule, $\tilde{V}_p'(\hat{\ell}) \to 0$ at both extremes. The incentive gradient is therefore bounded: it attains a finite maximum $\overline{V}_p = \max_{\hat{\ell}} \tilde{V}_p'(\hat{\ell})$ and a finite minimum $\underline{V}_p = \min_{\hat{\ell}} \tilde{V}_p'(\hat{\ell})$. When $\theta < -\overline{V}_p$, the objective is strictly decreasing in $\hat{\ell}$, driving the report to $\hat{p} = 0$; when $\theta > -\underline{V}_p$, the objective is strictly increasing, driving the report to $\hat{p} = 1$.

The asymmetry between models is sharpest near the boundaries. For the Brier score at p near 0:

$$\tilde{V}_p'(\hat{\ell}) = 2(p - \sigma(\hat{\ell}))\, \sigma(\hat{\ell})(1 - \sigma(\hat{\ell})). \tag{31}$$

The positive part of this gradient (the restoring force pulling the agent back from $\hat{p} = 0$) is maximized for $\sigma(\hat{\ell}) < p$, and scales as $\overline{V}_p = O(p^2)$ as $p \to 0$.[1] Boundary reports at $\hat{p} = 0$ therefore occur whenever $\theta < -\overline{V}_p$, which has probability

$$\Pr(\hat{p} = 0) = \Phi\left(\frac{-\overline{V}_p}{\sigma}\right) \xrightarrow{p \to 0} \Phi(0) = \frac{1}{2},$$

since $\overline{V}_p/\sigma \to 0$ for any fixed $\sigma > 0$. In words: as the true probability approaches 0, the restoring force in log-odds vanishes, and roughly half of all noise draws push the agent to the extreme report $\hat{p} = 0$, regardless of how small σ is.

Quantitative comparison for the Brier score. Table 2 illustrates the contrast. At $p = 0.05$ with $\sigma = 0.01$, the probability-space model produces essentially no boundary reports, while the log-odds model produces boundary reports for nearly half of all draws.

Table 2: Probability of boundary reports at $p = 0.05$ for the Brier score.

	Probability space	Log-odds
Threshold for boundary	$\lvert\theta\rvert > 2p = 0.1$	$\theta < -\overline{V}_p \approx -0.001$
$\sigma = 0.001$	$< 10^{-23}$	$\approx 16\%$
$\sigma = 0.01$	$< 10^{-23}$	$\approx 46\%$
$\sigma = 0.1$	$\approx 32\%$	$\approx 50\%$

The pattern generalizes beyond the Brier score. For any scoring rule with bounded payments $S(\hat{p}, x) \in [0, M]$, the expected score $V_p(\hat{p})$ is bounded, so $\tilde{V}_p(\hat{\ell}) = V_p(\sigma(\hat{\ell}))$ is bounded. Since $\tilde{V}_p$ is bounded while $\theta\hat{\ell}$ is linear in $\hat{\ell}$, the objective $\tilde{V}_p(\hat{\ell}) + \theta\hat{\ell}$ is eventually dominated by the noise term for large $|\hat{\ell}|$. The incentive gradient $\tilde{V}_p'$ is therefore bounded, and the same boundary-report phenomenon occurs.

Since the probability-space Taylor approximation performs well numerically (Section **??**) and the support issue can be handled by clipping $\hat{p}$ to $[0, 1]$, the remainder of the paper works exclusively in probability space.

[1] Writing $s = \sigma(\hat{\ell})$, the positive part is $2(p - s)s(1 - s)$ for $s \in (0, p)$. Substituting $s = pt$ for $t \in (0, 1)$ gives $2p^2\, t(1 - t)(1 - pt) \approx 2p^2\, t(1 - t)$ for small p, maximized at $t = 1/2$ with value $p^2/2$.

11 Wager extension

The limited liability constraint $S(\hat{p}, x) \geq 0$ models tournament-style platforms (e.g. Metaculus) where forecasters cannot lose money. In prediction markets, the agent puts up a stake $W > 0$ and can lose part or all of it. The payoff constraint becomes $S(\hat{p}, x) \geq -W$ for all $\hat{p} \in (0, 1)$ and $x \in \{0, 1\}$, and we impose a *participation constraint*: the agent's expected payment when reporting truthfully must be non-negative, $G(p) \geq 0$ for all $p \in (0, 1)$.

11.1 Reducing the wager constraint

Define $\bar{S}(\hat{p}, x) = S(\hat{p}, x) + W$. Then $\bar{S} \geq 0$, and $\bar{S}$ is a strictly proper scoring rule with entropy function $\bar{G}(p) = G(p) + W$. Since $\bar{G}'' = G''$, the limited liability analysis of Section 7.1 applies to $\bar{S}$: the constraint $\bar{S} \geq 0$ reduces to

$$\bar{G}(0) + \bar{G}'(0) \geq 0, \qquad \bar{G}(1) - \bar{G}'(1) \geq 0,$$

i.e.,

$$G(0) + G'(0) \geq -W, \qquad G(1) - G'(1) \geq -W. \tag{32}$$

As in Section 7, adding an affine function $a + bp$ to G does not change G'' or the loss term, but shifts expected payments. The principal minimizes payments by choosing a and b so that both constraints bind:

$$G(0) + G'(0) = -W, \qquad G(1) - G'(1) = -W. \tag{33}$$

11.2 Green's function representation with wager

Following the derivation in Section 7.3 with the modified boundary conditions (33), we integrate $G''(p) = g(p)$ twice. The first boundary condition gives $G(0) = -W - G'(0)$. Substituting into $G(p) = G(0) + G'(0)\,p + \int_0^p (p - t)\,g(t)\,dt$ and applying the second boundary condition yields:

$$G'(0) = -\int_0^1 t\,g(t)\,dt, \qquad G(0) = -W + \int_0^1 t\,g(t)\,dt.$$

These are identical to the limited liability case except for the $-W$ shift in $G(0)$. Assembling:

$$G(p) = \int_0^1 K(p, t)\,g(t)\,dt - W, \tag{34}$$

where $K(p, t)$ is the same Green's function kernel (17) as before.

11.3 Participation constraint

The participation constraint $G(p) \geq 0$ becomes $\Phi(p) \equiv \int_0^1 K(p, t)\,g(t)\,dt \geq W$ for all $p \in (0, 1)$. To determine where this constraint is tightest, we compute:

$$\Phi'(p) = \int_0^p (1 - t)\,g(t)\,dt - \int_p^1 t\,g(t)\,dt, \qquad \Phi''(p) = g(p) > 0.$$

Since Φ is strictly convex in p, its minimum is achieved at a unique interior point p^* satisfying $\Phi'(p^*) = 0$, i.e.,

$$\int_0^{p^*} (1 - t)\,g(t)\,dt = \int_{p^*}^1 t\,g(t)\,dt. \tag{35}$$

Therefore, it suffices to show that the participation constraint is satisfied at the minimum p^*:

$$\int_0^1 K(p^*, t)\, g(t)\, dt \geq W. \tag{36}$$

When $W = 0$, the constraint is automatically satisfied since $\Phi(p^*) > 0$ (because $K \geq 0$, $K \not\equiv 0$, $g > 0$). For $W > 0$, it imposes a lower bound on total curvature.

11.4 Principal's objective with wager

The principal's expected payment is $\mathbb{E}_f[G(p)] = \int_0^1 f(p)\, G(p)\, dp = \int_0^1 \phi(p)\, g(p)\, dp - W$, using (34) and the definition of ϕ (18). The $-W$ term reflects the principal's gain from collecting the agent's stake. The full objective is:

$$\int_0^1 \left[\frac{\sigma^2\, H(p)\, f(p)}{2\, g(p)^2} - \alpha\, \phi(p)\, g(p) \right] dp + \alpha W, \tag{37}$$

subject to the participation constraint (36) and $g > 0$. The integrand is identical to the limited liability objective (20); the only change is the constant αW and the additional constraint.

11.5 Lagrangian optimization

The participation constraint requires $\Phi(p) \equiv \int_0^1 K(p, t)\, g(t)\, dt \geq W$ for all $p \in (0, 1)$. Because Φ is strictly convex, it achieves its global minimum at a unique interior point p^*. Consequently, the infinite family of constraints reduces to a single worst-case binding condition at this minimum: $\Phi(p^*) \geq W$.

We introduce a single scalar Lagrange multiplier $\lambda \geq 0$ for this constraint. The Lagrangian is:

$$\mathcal{L} = \int_0^1 \left[\frac{\sigma^2\, H(p)\, f(p)}{2\, g(p)^2} - \alpha\, \phi(p)\, g(p) \right] dp + \alpha W + \lambda \left(\int_0^1 K(p^*, t)\, g(t)\, dt - W \right). \tag{38}$$

To optimize this pointwise, we merge the constraint integral into the main objective. By renaming the dummy variable t to p in the constraint term and factoring out $g(p)$, the Lagrangian becomes separable in g:

$$\mathcal{L} = \int_0^1 \left[\frac{\sigma^2\, H(p)\, f(p)}{2\, g(p)^2} - \Big(\alpha\, \phi(p) - \lambda\, K(p^*, p) \Big) g(p) \right] dp + W(\alpha - \lambda).$$

We define the *effective cost of curvature* as the term inside the parentheses:

$$\tilde{\phi}(p) \equiv \alpha\, \phi(p) - \lambda\, K(p^*, p). \tag{39}$$

The wager effectively subsidizes the principal's cost of paying the agent, reducing the baseline cost of curvature $\alpha\phi(p)$ by the discounted constraint penalty $\lambda K(p^*, p)$. Provided $\tilde{\phi}(p) > 0$, maximizing the integrand pointwise with respect to $g > 0$ yields the optimal curvature:

$$\boxed{G''(p)^*_W = \left(\frac{\sigma^2\, |H(p)|\, f(p)}{\tilde{\phi}(p)} \right)^{1/3},} \tag{40}$$

where the multiplier $\lambda \geq 0$ and the binding point p^* are determined simultaneously by the minimization of Φ and the complementary slackness condition:

$$\lambda \left(\int_0^1 K(p^*, t)\, g^*(t)\, dt - W \right) = 0, \qquad p^* = \arg\min_p \int_0^1 K(p, t)\, g^*(t)\, dt. \tag{41}$$

11.6 Interpretation

The wager reduces the effective cost of curvature from $\alpha\,\phi(p)$ to $\tilde{\phi}(p) = \alpha\,\phi(p) - \lambda\,K(p^*, p)$. Since $K \geq 0$ and $\lambda \geq 0$, we have $\tilde{\phi}(p) \leq \alpha\,\phi(p)$, so the optimal curvature is everywhere at least as large as under limited liability. This is intuitive: the agent's willingness to risk losing money allows the principal to offer a scoring rule with stronger incentives (higher curvature) without increasing net expected payments.

The reduction is not uniform across p: the kernel $K(p^*, p)$ is minimized at $p = p^*$ (with value $p^*(1-p^*)$) and increases toward the boundaries ($K(p^*, 0) = p^*$, $K(p^*, 1) = 1 - p^*$). The effective cost is therefore reduced *most* near $p = 0$ and $p = 1$, and *least* near p^*. This reflects the structure of the Green's function: curvature near the boundaries contributes most to the expected payment at the constraint-binding point p^*.

When $W = 0$, the participation constraint is slack, $\lambda = 0$, and (40) reduces to the limited liability solution (22).

11.7 Feasibility and maximum wager

For the solution (40) to be valid, we require $\tilde{\phi}(p) > 0$ for all $p \in (0, 1)$. This imposes:

$$\alpha\,\phi(p) > \lambda\,K(p^*, p) \quad \text{for all } p \in (0, 1).$$

Since $\phi(p) > 0$, this holds for λ sufficiently small. As W increases, λ must increase to satisfy the participation constraint, eventually violating $\tilde{\phi} > 0$ at some p. The critical point occurs when $\tilde{\phi}$ first reaches zero at the minimizer of $\alpha\,\phi(p) - \lambda\,K(p^*, p)$, beyond which the optimization problem has no interior solution.

11.8 Counterexample with wager

Take $u(d, x) = -(d - x)^2$ and $f \equiv 1$ as in the limited liability counterexample. Then $H(p) = -2$, $\phi(p) = (p^2 - p + 1)/2$. By symmetry of the problem (f uniform, $|H|$ constant, ϕ symmetric about $p = 1/2$), the optimal g^* is symmetric and the participation constraint binds at $p^* = 1/2$. The kernel at this point is:

$$K(1/2,\, p) = \begin{cases} (1-p)/2 & p \leq 1/2, \\ p/2 & p > 1/2, \end{cases}$$

which can be written as $K(1/2, p) = \max(p,\, 1-p)/2$. The effective cost becomes:

$$\tilde{\phi}(p) = \frac{\alpha(p^2 - p + 1) - \lambda \max(p,\, 1-p)}{2}.$$

The optimal curvature is:

$$g_W^*(p) = \left(\frac{4\sigma^2}{\alpha(p^2 - p + 1) - \lambda \max(p,\, 1-p)} \right)^{1/3}, \tag{42}$$

where $\lambda \geq 0$ is determined by $\int_0^1 K(1/2,\, t)\, g_W^*(t)\, dt = W$ (if the constraint binds) or $\lambda = 0$ (if it does not).

Feasibility. By symmetry it suffices to consider $p \in [0, 1/2]$, where $\tilde{\phi}(p) = [\alpha p^2 + (\alpha - \lambda)(1 - p)]/2$. This is minimized at $p = (\alpha - \lambda)/(2\alpha)$, with minimum value $(\alpha - \lambda)(3\alpha + \lambda)/(8\alpha)$. Therefore $\lambda < \alpha$ is required. As $\lambda \to \alpha$, the minimizer of $\tilde{\phi}$ approaches $p = 0$ (and by symmetry $p = 1$), and

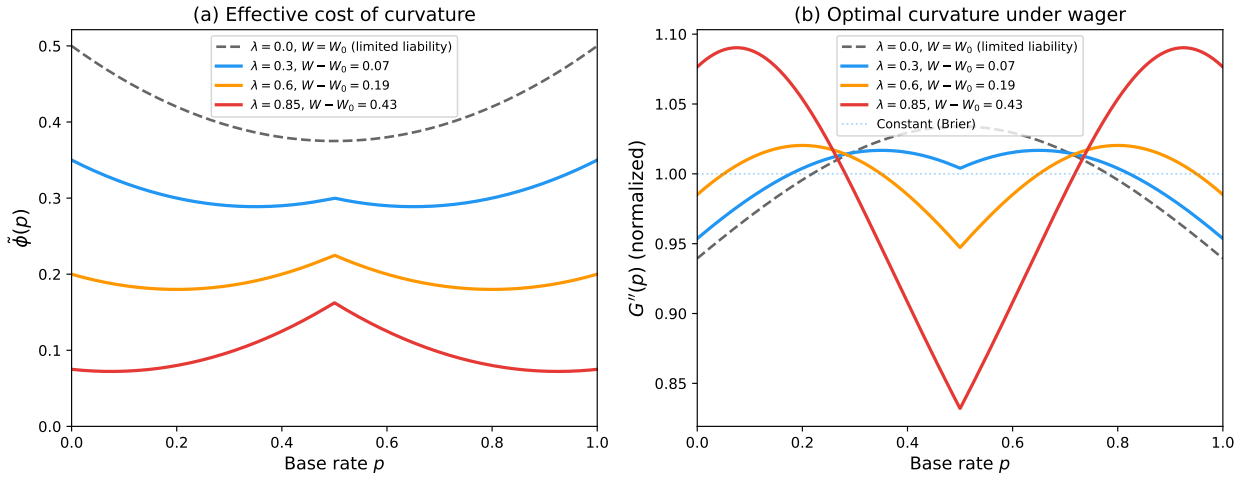

Figure 2: Effect of the wager on the counterexample ($u = -(d-x)^2$, $f \equiv 1$, $\alpha = \sigma^2 = 1$). Here $W_0 = \Phi(p^*)\big|_{\lambda=0}$ is the agent's minimum expected payoff under the limited liability solution; wagers $W \leq W_0$ are automatically satisfied and do not affect the optimum. (a) The effective cost of curvature $\tilde{\phi}(p)$ decreases as λ increases, with the largest reduction near the boundaries where $K(1/2, p)$ is largest. (b) The optimal curvature profile (normalized by its mean) shifts from center-peaked under limited liability to boundary-peaked under a large wager.

curvature diverges near the boundaries. The maximum feasible wager $W_{\max}$ is the value at which $\lambda = \alpha$.

Effect of the wager. As W increases from 0, the multiplier λ increases, reducing $\tilde{\phi}$ and raising curvature everywhere. The reduction in $\tilde{\phi}$ is largest near the boundaries (where $K(1/2, p)$ is largest) and smallest at $p = 1/2$ (where $K(1/2, p)$ is smallest). While the baseline curvature under limited liability is highest at $p = 1/2$ (where ϕ is smallest), the wager preferentially increases curvature near the boundaries, making the curvature profile more uniform. For large W, curvature diverges near the boundaries (as $\lambda \to \alpha$), in contrast to the limited liability solution where curvature peaks at $p = 1/2$.

The intuition is as follows. Under limited liability, the principal pays for curvature through the expected scoring rule payment, and this cost is mediated by $\phi(p)$: curvature at $p = 1/2$ is cheapest because it contributes least to expected payments. The wager introduces a second channel. The agent's stake is at risk, and the amount at risk depends on the Green's function kernel $K(p^*, p)$, which is largest near the boundaries. Curvature near the boundaries therefore "pays for itself" through the agent's wager exposure, reducing the principal's net cost there. As W increases, this boundary subsidy grows until it dominates the baseline cost profile, continuously rotating the optimal curvature from center-peaked through approximately uniform to boundary-peaked. At moderate wager levels, the two effects roughly balance, producing a nearly flat curvature profile. Figure 2 illustrates this progression.

12 Tail probability of errors

The variance formula (6) characterizes the average squared reporting error under a given scoring rule. For applications where the principal is concerned with the risk of large errors—rather than their average magnitude—the tail probability $\Pr(|\hat{p} - p| > \Delta)$ for a fixed threshold $\Delta > 0$ is a more

informative quantity. This section derives the tail probability under the Gaussian approximation, analyzes how it varies across scoring rules and base rates, and formulates an alternative optimization problem in which the principal minimizes the expected tail probability rather than the expected squared loss.

12.1 Tail probability under the Gaussian approximation

Under the probability-space random utility model (Section 5), the reporting error is approximately $\hat{p} - p \approx \theta / G''(p)$, where $\theta \sim N(0, \sigma^2)$. Therefore $\hat{p} - p$ is approximately $N(0,\, \sigma^2/[G''(p)]^2)$. For any threshold $\Delta > 0$:

$$\Pr(|\hat{p} - p| > \Delta) \approx \Pr\big(|\theta| > \Delta\, G''(p)\big) = \Pr\left(|Z| > \frac{\Delta\, G''(p)}{\sigma}\right),$$

where $Z = \theta/\sigma \sim N(0,1)$. By the symmetry of the standard normal, $\Pr(|Z| > z) = 2\,\bar{\Phi}(z)$, so

$$\Pr(|\hat{p} - p| > \Delta) = 2\,\bar{\Phi}\left(\frac{\Delta\, G''(p)}{\sigma}\right), \tag{43}$$

where $\bar{\Phi} = 1 - \Phi$ is the standard normal survival function. The tail probability depends on the scoring rule only through $G''(p)$ at the base rate p, via the signal-to-noise ratio

$$z(p) \equiv \frac{\Delta\, G''(p)}{\sigma}, \tag{44}$$

which measures the error threshold in units of the reporting standard deviation. Larger curvature $G''(p)$ increases $z(p)$, making large errors less likely.

12.2 Asymptotic behavior and exponential sensitivity

The exact survival function $\bar{\Phi}(z) = \int_z^\infty \varphi(t)\, dt$ has no closed form, but for large z the integrand decays rapidly away from the lower limit, so the integral is dominated by the contribution near $t = z$. The density drops by a factor of e^{-1} over a width of approximately $1/z$, giving the classical approximation known as Mill's ratio: $\bar{\Phi}(z) \sim \varphi(z)/z$. Substituting into (43) and writing out $\varphi(z) = (2\pi)^{-1/2}\, e^{-z^2/2}$:

$$\Pr(|\hat{p} - p| > \Delta) \;\approx\; \frac{2\sigma}{\Delta\, G''(p)\,\sqrt{2\pi}}\; \exp\left(-\frac{\Delta^2\,[G''(p)]^2}{2\sigma^2}\right). \tag{45}$$

The tail probability decays *exponentially* in $[G''(p)]^2$, in contrast to the variance $\sigma^2/[G''(p)]^2$, which decays only polynomially. Doubling the curvature halves the variance but squares the exponent in the tail bound, producing an exponentially larger reduction in tail probability. This distinction is economically significant: when the principal is concerned with catastrophic errors (large Δ), the curvature profile $G''(p)$ has a much stronger effect than the variance analysis alone would suggest.

For two scoring rules with curvatures $g_1(p)$ and $g_2(p)$ at a fixed base rate p, the ratio of tail probabilities is approximately:

$$\frac{\Pr(|\hat{p} - p| > \Delta;\, g_1)}{\Pr(|\hat{p} - p| > \Delta;\, g_2)} \;\approx\; \frac{g_2}{g_1}\, \exp\left(-\frac{\Delta^2\,(g_1^2 - g_2^2)}{2\sigma^2}\right). \tag{46}$$

When $g_1 > g_2$, the exponent is negative and grows in magnitude as Δ^2, so the higher-curvature rule's advantage grows exponentially as the error threshold increases.

12.3 Tail probability under the optimal curvature

Substituting the optimal curvature (22) into (44):

$$z^*(p) = \frac{\Delta}{\sigma}\left(\frac{\sigma^2\,|H(p)|\,f(p)}{\alpha\,\phi(p)}\right)^{1/3} = \Delta\left(\frac{|H(p)|\,f(p)}{\sigma\,\alpha\,\phi(p)}\right)^{1/3}. \tag{47}$$

The tail probability under the variance-optimal rule is therefore:

$$\Pr(|\hat{p} - p| > \Delta \mid p) = 2\,\bar{\Phi}(z^*(p)). \tag{48}$$

This is smallest (errors rarest) where $z^*(p)$ is largest, i.e., where $|H(p)|\,f(p)/\phi(p)$ is largest. The tail probability varies across base rates even under the optimal rule: the variance-optimal curvature tolerates higher tail probabilities at base rates where curvature is expensive ($\phi(p)$ large) and achieves lower tail probabilities where curvature is cheap ($\phi(p)$ small).

12.4 Effect of clipping

Under the clipped model (Section 7), the agent's actual report is $\hat{p}_c = \max(0, \min(1, \hat{p}_{\text{exact}}))$. Since $\hat{p}_c \in [0, 1]$ and $p \in (0, 1)$, the clipped error satisfies $|\hat{p}_c - p| \leq \max(p,\, 1 - p) < 1$. In particular:

$$\Pr(|\hat{p}_c - p| > \Delta) = 0 \quad \text{for all } \Delta \geq \max(p,\, 1 - p). \tag{49}$$

This hard upper bound does not exist for the unconstrained error $\hat{p}_{\text{exact}} - p$, which has unbounded Gaussian tails. For $\Delta < \min(p, 1 - p)$, clipping does not affect the tail probability—errors of magnitude Δ occur entirely in the interior of $[0, 1]$, so $\hat{p}_c = \hat{p}_{\text{exact}}$ whenever $|\hat{p}_{\text{exact}} - p| \leq \Delta$. For intermediate thresholds $\min(p, 1 - p) \leq \Delta < \max(p, 1 - p)$, clipping eliminates tail probability on one side (the nearer boundary) while leaving the other side unchanged. In summary:

$$\Pr(|\hat{p}_c - p| > \Delta) \leq \Pr(|\hat{p}_{\text{exact}} - p| > \Delta) = 2\,\bar{\Phi}\left(\frac{\Delta\,G''(p)}{\sigma}\right),$$

with strict inequality when $\Delta \geq \min(p, 1 - p)$. The Gaussian formula (43) is therefore an upper bound on the true tail probability under clipping.

12.5 Tail-optimal curvature

We now formulate an alternative optimization problem in which the principal minimizes the expected tail probability of the reporting error, rather than the expected squared loss. The objective is:

$$\min_{g>0} \int_0^1 \left[2\,\bar{\Phi}\left(\frac{\Delta\,g(p)}{\sigma}\right) f(p) + \alpha\,\phi(p)\,g(p)\right] dp. \tag{50}$$

The first term is the prior-weighted probability that the error exceeds Δ; the second is the cost of expected payments, identical to (20). The integrand is separable in g, so we optimize pointwise.

First-order condition. At each p, differentiate the integrand with respect to g:

$$-\frac{2\Delta}{\sigma}\,\varphi\left(\frac{\Delta\,g}{\sigma}\right) f(p) + \alpha\,\phi(p) = 0,$$

where $\varphi(z) = (2\pi)^{-1/2}\, e^{-z^2/2}$ is the standard normal density. Substituting $z = \Delta\, g/\sigma$:

$$\varphi(z) = \frac{\alpha\, \sigma\, \phi(p)}{2\Delta\, f(p)}. \tag{51}$$

Since $\varphi(z) = (2\pi)^{-1/2}\, e^{-z^2/2}$, solving for z:

$$z^2 = 2\ln\left(\frac{\Delta\, f(p)}{\alpha\, \sigma\, \phi(p)}\sqrt{\frac{2}{\pi}}\right). \tag{52}$$

A real solution exists when the argument of the logarithm exceeds 1, i.e., when the marginal benefit of curvature (measured by $\Delta\, f(p)$) exceeds the marginal cost (measured by $\alpha\, \sigma\, \phi(p)$) by a factor of $\sqrt{\pi/2}$.

Second-order condition. The second derivative of the integrand with respect to g is:

$$\frac{2\Delta^2}{\sigma^2}\, z\, \varphi(z)\, f(p) > 0 \quad \text{for } z > 0,$$

so the integrand is strictly convex in g for $g > 0$, confirming the first-order condition gives the unique minimum.

Tail-optimal curvature. The solution is:

$$\boxed{G''_{\text{tail}}(p) = \frac{\sigma}{\Delta}\sqrt{2\ln\left(\sqrt{\frac{2}{\pi}}\,\frac{\Delta\, f(p)}{\alpha\, \sigma\, \phi(p)}\right)}.} \tag{53}$$

12.6 Comparison of curvature allocation principles

Power law versus logarithm. The variance-optimal curvature scales as a power law in the benefit-to-cost ratio $|H(p)|\, f(p)/\phi(p)$:

$$G''_{\text{var}}(p) \propto \left(\frac{|H(p)|\, f(p)}{\phi(p)}\right)^{1/3}.$$

The tail-optimal curvature scales as the square root of a logarithm:

$$G''_{\text{tail}}(p) \propto \sqrt{\ln\left(\frac{f(p)}{\phi(p)}\right)}.$$

The logarithmic dependence is much weaker: the tail-optimal rule distributes curvature far more uniformly across base rates than the variance-optimal rule. The intuition is that tail probability decays exponentially in $[G''(p)]^2$, so the marginal benefit of additional curvature diminishes very rapidly once $z(p)$ is moderately large. Conversely, regions of low curvature contribute disproportionately to the expected tail probability, since $\bar{\Phi}(z)$ is large when z is small. The tail-optimal rule therefore avoids extreme curvature variation: it raises curvature where it is low (at high marginal benefit) and reduces it where it is already high (at low marginal benefit).

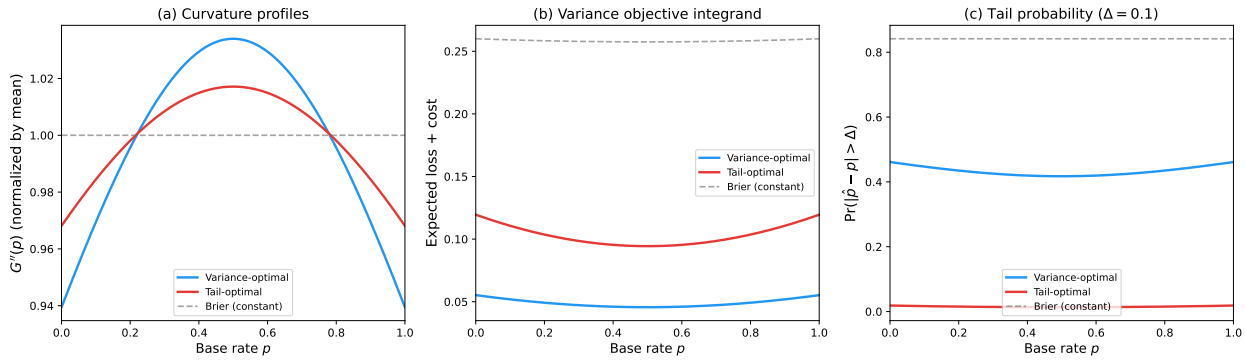

Figure 3: Comparison of variance-optimal and tail-optimal scoring rules for the counterexample ($u(d, x) = -(d - x)^2$, $f \equiv 1$, $\sigma = 1$, $\alpha = 0.01$, $\Delta = 0.1$). (a) Curvature profiles normalized by their mean: the variance-optimal rule concentrates curvature at $p = 1/2$ where it is cheapest, while the tail-optimal rule distributes curvature more uniformly. (b) The variance objective integrand $\sigma^2 |H(p)|\, f(p)/[2\, g(p)^2] + \alpha\, \phi(p)\, g(p)$: the variance-optimal rule achieves the lowest combined loss plus payment cost at every base rate. (c) Tail probability $\Pr(|\hat{p} - p| > \Delta)$: the tail-optimal rule achieves lower and more uniform tail probability. Each rule is optimal according to its own objective.

Dependence on the threshold. The variance-optimal curvature does not involve the threshold Δ. The tail-optimal curvature, on the other hand, depends on Δ both in scale and shape: increasing Δ increases z^* at every p, but the increase is larger where curvature was already low, slightly compressing the curvature variation. In the limit $\Delta \to 0$ (concern about any error), the tail-optimal $z \to 0$ and the approximation breaks down. In the limit $\Delta \to \infty$ (concern only about extreme errors), the logarithmic term dominates and the tail-optimal curvature becomes approximately uniform.

Role of the principal's sensitivity. The variance-optimal curvature depends on $|H(p)|$, the principal's sensitivity to errors at base rate p, because the expected loss weights squared errors by $|H(p)|$. The tail-optimal curvature as formulated in (53) does not involve $|H(p)|$, because the tail probability $\Pr(|\hat{p} - p| > \Delta)$ counts only whether the error exceeds the threshold, not how costly it is.

13 Robustness of the noise specification

The results in Sections 5–7 rest on two modeling choices: the linear noise specification $\varepsilon(\hat{p}) = \theta\hat{p}$ and the Gaussian distribution of θ. This section shows that neither is essential for the qualitative conclusions.

13.1 General noise functions

Replace the linear specification $\varepsilon(\hat{p}) = \theta\hat{p}$ with $\varepsilon(\hat{p}) = \theta \cdot m(\hat{p})$, where $m\colon (0, 1) \to \mathbb{R}$ is a smooth, strictly increasing function. The agent maximizes

$$V_p(\hat{p}) + \theta \cdot m(\hat{p}),$$

with first-order condition

$$V_p'(\hat{p}) + \theta \cdot m'(\hat{p}) = 0. \tag{54}$$

A first-order Taylor expansion of the left-hand side around $\hat{p} = p$ gives

$$V_p''(p)\,(\hat{p} - p) + \theta \cdot m'(p) \approx 0,$$

so that

$$\hat{p} - p \approx -\frac{\theta \cdot m'(p)}{V_p''(p)} = \frac{\theta \cdot m'(p)}{G''(p)}, \tag{55}$$

where the last equality uses $V_p''(p) = -G''(p)$ (Remark 1). The reporting variance becomes

$$\mathrm{Var}(\hat{p} - p) \approx \frac{\sigma^2\,[m'(p)]^2}{[G''(p)]^2}. \tag{56}$$

The factor $[m'(p)]^2$ depends on the noise specification and the base rate, but *not* on the scoring rule. It enters the principal's expected loss (8) as a multiplicative term:

$$\mathbb{E}[L \mid p] \approx -\frac{\sigma^2\,[m'(p)]^2\,H(p)}{2\,[G''(p)]^2}.$$

Since the additional factor $[m'(p)]^2$ is a fixed function of p, it does not affect the separability of the objective (20) in g. The pointwise optimization proceeds exactly as in Section 7.4, with $|H(p)|\,f(p)$ replaced by $|H(p)|\,f(p)\,[m'(p)]^2$.

Proposition 2 (General noise optimum)**.** *Under the noise specification $\varepsilon(\hat{p}) = \theta \cdot m(\hat{p})$ with $\mathbb{E}[\theta] = 0$ and $\mathrm{Var}(\theta) = \sigma^2$, the optimal curvature under limited liability is*

$$G''(p)^* = \left(\frac{\sigma^2\,|H(p)|\,f(p)\,[m'(p)]^2}{\alpha\,\phi(p)}\right)^{1/3}, \tag{57}$$

where $\phi(p)$ is the cost of curvature (18)*. Setting $m(\hat{p}) = \hat{p}$ (i.e., $m'(p) = 1$) recovers* (22)*.*

The qualitative interpretation is unchanged: the optimal rule concentrates curvature where errors are costly ($|H(p)|$ large), base rates likely ($f(p)$ large), noise sensitivity high ($m'(p)$ large), and curvature cheap ($\phi(p)$ small). In particular, the structure of the optimization—separable objective, cubic pointwise problem, Green's function representation—is entirely preserved. The linear specification $m(\hat{p}) = \hat{p}$ is the canonical simplest case of this family.

13.2 Relaxing the Gaussian assumption

The core optimization results (Sections 6–7) use only two properties of θ:

1. $\mathbb{E}[\theta] = 0$ (unbiased noise), which ensures $\mathbb{E}[\hat{p} - p] \approx 0$;
2. $\mathbb{E}[\theta^2] = \sigma^2$ (finite variance), which determines the expected squared error via (6).

No higher moments or distributional shape enters the derivation of the optimal curvature (22) or its generalization (57). The Taylor approximation $\hat{p} - p \approx \theta/G''(p)$ is valid under any distribution of θ with finite variance, provided $\sigma/G''(p)$ is small enough that higher-order terms are negligible.

Therefore, the main results can be stated under the weaker moment conditions $\mathbb{E}[\theta] = 0$ and $\mathbb{E}[\theta^2] = \sigma^2$, without requiring Gaussianity.

The Gaussian assumption becomes necessary only in Section 12, where the explicit distributional form $\hat{p} - p \sim N(0, \sigma^2/[G''(p)]^2)$ is used to compute tail probabilities via the normal survival function $\bar{\Phi}$. For non-Gaussian θ, the tail analysis would require either (i) the specific distribution of θ, or (ii) distribution-free bounds such as Chebyshev's inequality, which gives the weaker bound

$$\Pr(|\hat{p} - p| > \Delta) \leq \frac{\sigma^2}{[G''(p)]^2\, \Delta^2}.$$

This preserves the qualitative dependence on $G''(p)$ but replaces the exponential tail decay of the Gaussian case with a polynomial bound.